Context Spatial Awareness Remote Sensing Image Change Detection Network Based on Graph and Convolution Interaction

Xinyang Song, Zhen Hua, and Jinjiang Li

***Abstract*— Remote sensing (RS) images are characterized by high dimensionality, complex textures, and large scales. Traditional convolutional neural network (CNN) methods may overlook spatial relationships and contextual information among pixels when dealing with RS data. Therefore, graph convolutional networks (GCNs) have emerged as a promising solution. In this article, we propose a contextual spatial awareness remote sensing image change detection network based on the interaction between graphs and convolutions (CSAGC). We aim to enhance the handling of contextual information by introducing multiple augmentation modules. In CSAGC, we propose a high-performance encoder called Congraph that integrates a CNN and a graph neural network (GNN). By preserving the respective features of both branches, we effectively fuse local detailed features and global positional features, achieving superior feature extraction capabilities. Additionally, we design two modules to facilitate the integration of multiscale spatial information: contextual spatial awareness module (CSAM) and spatial integration module (SIM). CSAM, a crucial module connecting the encoder and decoder, jointly explores contextual features using the current feature branch and high-low level feature branches, leveraging spatial positional information for better content acquisition. SIM, located in the decoder module, aims to integrate the multiscale information outputted by CSAM, complementing the contextual information and improving the overall network's ability to capture spatial contextual information. We conducted extensive experiments on three datasets, namely LEVIR-CD, WHU-CD, and GZ-CD. The experimental results demonstrate that CSAGC exhibits excellent performance, achieving significant performance improvements compared to state-of-the-art (SOTA) methods.**

***Index Terms*— Attention mechanism, convolutional neural network (CNN), graph convolution, remote sensing (RS) change detection (CD).**

I. Introduction

With the advancement of remote sensing (RS) image technology, the task of RS image change detection (CD) has made significant progress and gradually become a hot topic in the field of RS. RS CD involves extracting change information from high-resolution RS images collected at different times within the same area using various image processing techniques and pattern recognition methods, aiming to study the surface changes in that area [1]. Depending on the nature of the target being detected, RS CD can be applied to a wide range of scenarios. Currently, successful applications of RSCD include urban management [2], damage assessment [3], deforestation monitoring [4], environmental monitoring [5], agricultural land changes [6], and others. Due to the varying characteristics of the target objects, the performance of CD methods may differ.

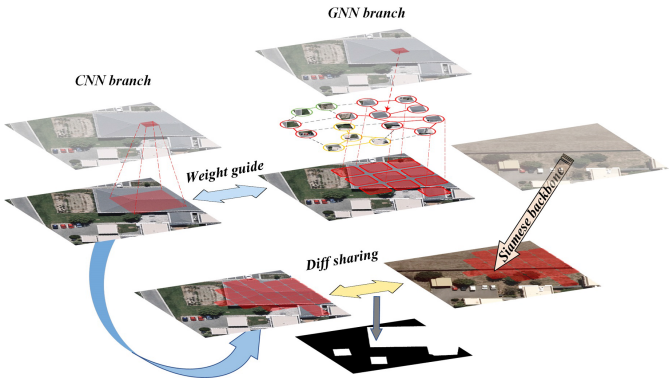

Fig. 1. Schematic description of the backbone for interacting with GNNs using CNNs.

RS CD can be viewed as a multiimage binary segmentation task. However, CD requires considering both the common and differing features among multiple images. Thus, any error in individual images can lead to differences in the final results. Additionally, factors such as variations in illumination angles, shadows, and reflectance of ground objects across different time periods can significantly impact the performance of CD. These factors contribute to the increased difficulty of CD compared to segmentation tasks.

In early traditional CD methods, due to the availability of lower resolution RS data, conventional image analysis techniques were often employed to calculate difference images. For example, CD methods based on clustering, such as k-means clustering [7], [8], or threshold-based methods [9].

Principal component analysis (PCA) [7] and change vector analysis (CVA) [10] have also demonstrated several

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Xinyang Song and Jinjiang Li are with the School of Information and Electronic Engineering, Institute of Network Technology (ICT), Shandong Technology and Business University, Yantai 264005, China.

Zhen Hua is with the School of Computer Science and Technology, Shandong Technology and Business University, Yantai 264005, China (e-mail: huazhen@sdtbu.edu.cn).

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advantages in RSCD. They effectively handle RS data by reducing dimensions, visualizing data, extracting features, and capturing major patterns of change without the need for prelabeled training data. These techniques play a significant role in efficiently analyzing RS data and identifying prominent change patterns.

However, when dealing with complex RS image data, traditional statistical methods may struggle to capture nonlinear relationships and local features effectively. Therefore, the introduction of convolutional neural networks (CNNs) in deep learning (DL)-based CD methods [11] can address these issues more effectively. DL methods leverage CNNs to automatically learn feature representations, handle large-scale datasets efficiently, and utilize the local receptive fields (RFs) and parameter sharing of convolutional layers to improve the accuracy and robustness of RS CD. The success of DL extends beyond CNNs alone, as generative adversarial networks (GANs) [12], [13] and recurrent neural networks (RNNs) [14], [15] have also been applied in CD tasks. These powerful DL-based methods rely on their excellent modeling and feature extraction capabilities, requiring minimal additional design to achieve recognition of high-level information features and end-to-end CD in RS data.

With the continuous improvement of high-performance CNNs and the increasing quality of high-resolution RS images, there are higher expectations for the feature extraction capabilities of bitemporal images in CD tasks. Many studies have improved CD methods [16], [17] based on excellent segmentation models [18], [19], achieving outstanding results. However, traditional convolutions are limited by their RFs and can only capture local contextual information. To enable CNNs to better capture long-range dependencies in image processing, researchers have started to stack more convolutional layers [20], [21], [22], [23] or use dilated convolutions to expand the RF [24]. Furthermore, in order to further improve the performance of CNNs, researchers have explored the use of attention mechanisms [21], [22], [25], [26], [27], [28]. The introduction of attention mechanisms in CD tasks allows CNN networks to focus more on key regions by learning importance weights, thereby enhancing the attention given to change information.

RS images possess high dimensionality, complex textures, and large scales, making it challenging for traditional image processing methods to handle such data effectively. With the powerful contextual modeling capability demonstrated by transformers (TRs), CD networks based on TR have been continuously developed and achieved remarkable performance. TR can capture longer range dependencies, compensating for the global perspective that traditional convolutions and attention mechanisms may lack. In CD tasks, numerous methods [29], [30], [31], [32], [33] have emerged with outstanding performance, demonstrating the superior ability of TR in handling complex images for CD.

In addition to the sequential structure of vision transformers (ViTs) for handling complex RS images, graph convolutional networks (GCNs) have also demonstrated excellent capabilities in recent RS research [34], [35], [36], [37], [38], [39]. By propagating and aggregating information between nodes, GCN can utilize the neighboring information of nodes to update their feature representations. This enables GCN to better capture both local and global spatial relationships in RS images, thereby enhancing the processing and analysis capabilities of RS data. GCN leverages the connectivity relationships between nodes to model the characteristics of RS data, further extracting spatial contextual information from the image. This ability of graph convolution plays a crucial role in tasks such as RS image classification, object detection, and land cover recognition, offering significant potential for the development of RS image processing and analysis.

In this article, we have introduced a contextual spatial awareness remote sensing image change detection network based on the interaction between graphs and convolutions (CSAGC). Within this network, we integrate a high-performance encoder named Congraph. Congraph effectively harnesses the synergy between CNN and GNN, seamlessly blending local fine-grained features with global contextual information. This integration substantially augments its feature extraction capabilities. Congraph specifically underscores the distinct advantages of GNNs in managing global information and establishing connections, thus complementing the CNN's primary focus on capturing local features. This harmonious fusion renders Congraph a high-performance encoder. Fig. 1 illustrates the concept of Congraph. To facilitate the fusion of spatial information from different scales, we introduce the contextual spatial awareness module (CSAM). This module considers up to three adjacent scales to provide comprehensive global assistance for the current features. Additionally, in the decoder, we employ residual dilated convolutions to construct the spatial integration module (SIM) for contextual information aggregation. The overall network architecture is depicted in Fig. 2. Under this network structure, our CSAGC demonstrates significant advantages compared to state-of-the-art (SOTA) methods.

Our main contribution can be summarized as follows.

1) We propose Congraph, a high-performance hybrid backbone that leverages multilayer interaction between CNN and GNN branches to exchange and fuse information, thereby capturing diverse and more comprehensive global information and local features.
2) We introduce a CSAM that integrates spatial information from current-scale features and cross-scale features, enabling a more comprehensive representation of feature information.
3) In our framework, CSAGC serves as a high-performance network for CD tasks, employing an encoder–decoder architecture. Congraph acts as the encoder, while the decoder consists of a multilayer aggregation of SIM formed by dilation convolutions. Within this framework, our CSAGC achieves significant performance improvements compared to SOTA methods.

The rest of this article is organized as follows. Section II describes the nontransformer-based approach with the transformer based CD approach under the DL approach. In Section III, we describe the design ideas in our method in detail. In Section IV, we summarize and analyze a large

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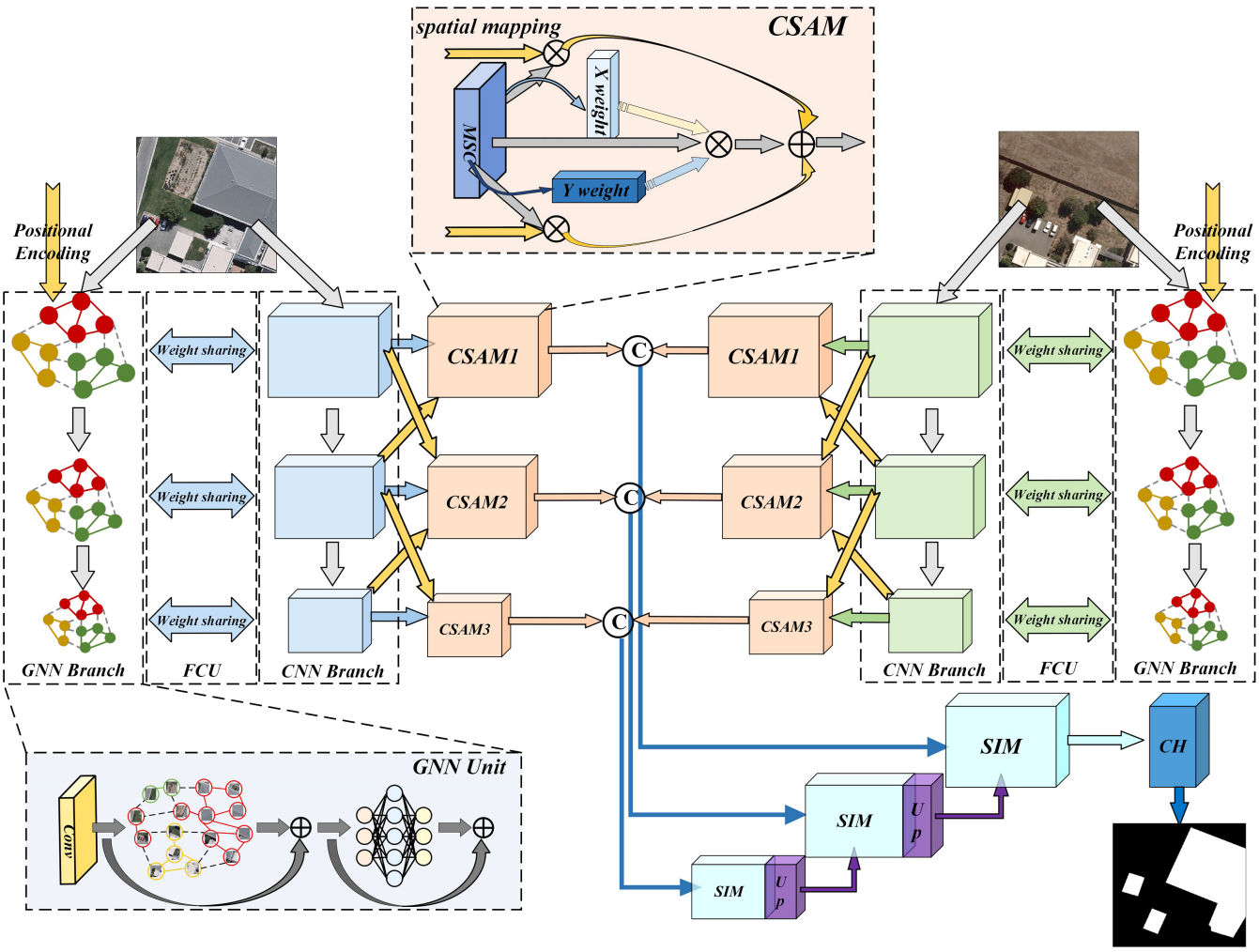

Fig. 2. Network structure of CSAGC. It consists of Congraph, an interactive backbone that combines CNN and GNN, as well as CSAM for modeling contextual spatial information and SIM for integrating multiscale features.

number of experimental reports. Finally, conclusions are drawn in Section V.

II. Related Work

DL has achieved remarkable success in the field of computer vision. In recent years, there has been a significant improvement in the performance of CD methods based on DL. Methods based on CNN and attention mechanisms have been continuously developed, innovated, and improved to tackle the increasingly complex representations of RS images. With the advancements in RS image technology, the complexity of images has also increased. To better handle these complex images, a large number of methods based on GCN and TR have been developed, demonstrating outstanding performance. In the following sections, we will introduce methods based on CNN, GCN, and TR separately.

A. CNN-Based CD Method

CNN methods have proven to be indispensable in various fields. As the development and innovation of CNNs continue in the field of segmentation tasks, researchers have explored the use of excellent segmentation models for analyzing differential features in RS CD tasks. Daudt et al. [17] utilized a Siamese network under the Unet framework and investigated different input modes for bitemporal images. They employed fully convolutional neural networks (FCNs) with three different processing approaches, providing a solid foundation for subsequent research. Similarly, other works such as [40] have also demonstrated the successful application of improved UNet++ models in RS CD tasks.

To enhance the recognition capability of features, attention mechanisms have gradually become an integral part of many DL-based methods. With the widespread use of attention mechanisms in various domains, numerous attention-based methods have emerged in the field of CD tasks. These methods enhance the feature weights by incorporating attention mechanisms into CNN architectures, allowing for better extraction of informative features. For example, Fang et al. [23] fused feature information at each layer of a dense network using channel attention, resulting in significant performance improvements compared to CNN alone. Chen and Shi [25] employed self-attention to extract crucial feature information, enabling better modeling of spatiotemporal relationships, and integrated it into a feature extraction process. In addition to emphasizing feature information, researchers also focus on

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enlarging the RF in traditional CNN methods. To address this, Song et al. [40] replaced traditional convolutions with dilated convolutions and variable convolutions to increase the RF, achieving better contextual modeling capabilities.

Under CNN-based methods, network performance improvement mostly relies on techniques such as multiscale information fusion [23] and deep supervision [22]. Although these methods can bring notable performance gains, they often come with increased parameter complexity and computational costs. Moreover, due to the inability to establish long-range dependencies effectively, deep networks still suffer from limitations in terms of RF size.

Therefore, in recent years, many studies have focused on how to establish global dependencies more effectively. Researchers aim to capture long-range relationships and address the limited RF issue in deep networks. Various approaches have been explored, including attention mechanisms [41], [42], GCNs [43], [44], and Transformer models [45], [46]. These techniques aim to enhance the network's ability to capture global dependencies and leverage contextual information more effectively. By incorporating these advancements, researchers seek to overcome the limitations of traditional CNNs and improve the performance of CD in RS images.

B. Transformer-Based Method

TR [47] have demonstrated powerful performance in various computer vision domains. To handle increasingly complex RS images and capture a larger RF, CD methods have started incorporating TR. Two types of CD structures based on TRs have emerged.

The first type involves using a separate feature extraction backbone to extract features and then utilizing a TR module to establish the connection between dual-temporal images for extracting feature difference information [6], [29], [33], [48]. For instance, Chen et al. [29] employed a ResNet18 as the feature extraction backbone and utilized a TR module for modeling feature differences, achieving superior performance compared to previous deep CNN methods.

The second type of CD method based on TR treats TR as the feature extraction backbone and directly extracts feature difference information during the feature extraction process of dual-temporal images. Multiple scales of feature difference information are progressively propagated backward for the final feature difference prediction [30], [31]. For example, Zhang et al. [31] proposed a pure TR model for CD, utilizing an encoder–decoder structure based on the Unet architecture for CD tasks.

While ViT have shown excellent performance in tasks such as image classification and object detection, there are certain application scenarios, such as RS CD, where complex spatial dependencies exist between pixels. In such cases, dividing the image into blocks and aggregating them using self-attention mechanisms may not capture the global dependencies between pixels effectively. Therefore, alternative model structures, such as Vision graph neural networks (GNNs), are needed to handle the relationships between pixels. As a result, we explore the utilization of both CNN and GNN in the backbone to obtain more comprehensive feature information.

C. GCN-Based CD Method

As RS image quality continues to improve, they exhibit more complex textures, which makes it increasingly challenging for traditional CNN methods to handle such data. In modeling complex images, both the positional information and global contextual information are crucial for extracting detailed local features and global characteristics. Neglecting these aspects may compromise the integrity and continuity of the results.

GCNs offer a technique to perform convolutions on graph-structured data, which is different from the traditional pixel-based convolutions. In this context, an image is viewed as a graph, where each pixel corresponds to a node in the graph, and the relationships between pixels correspond to the edges in the graph. By applying convolutions on the nodes and edges of the graph, spatial features and local structures within the image can be effectively captured. In the field of CD methods, there have been various attempts involving GCN [49]. However, most of these methods utilize GCN as a backbone network and employ unsupervised or semisupervised approaches for training.

In the field of RS, many methods have attempted to improve GCN by combining it with CNN to enable CNN to benefit from the global attention capabilities of GCN [34], [35], [36], [38], [39]. For instance, Liu et al. [34] proposed a heterogeneous deep network called CNN-enhanced GCN (CEGCN). It leverages the strengths of both CNN and GCN by generating complementary feature maps at both pixel and superpixel levels. Similarly, in works like [35], [36], a similar combination of CNN and GCN is employed. These methods either overlay GCN on CNN or fuse them in a single manner, resulting in the neglect of many intermediate feature points.

We aim to fully exploit the distinctive feature information brought by CNN and GNN. In this article, we fuse features from multiple layers of GNN and CNN, designing a hybrid backbone with multilayer interactions between GNN and CNN. This design allows for the complementary utilization of their respective features, enabling the maximum preservation of local and global feature perception.

III. Methodology

In this article, we propose a high-performance CSAGC network, as illustrated in Fig. 2. Our network consists of Congraph, an interactive backbone that combines CNN and GNN, as well as CSAM for modeling contextual spatial information and SIM for integrating multiscale features. Congraph serves as the encoder, while SIM functions as the decoder. To provide a more intuitive understanding of our method's workflow, we present the inference details in Algorithm 1.

A. Congraph Encoder

1) Overview: In the processing of high-resolution RS images, images with high dimensionality and complex textures often require powerful feature extraction capabilities.

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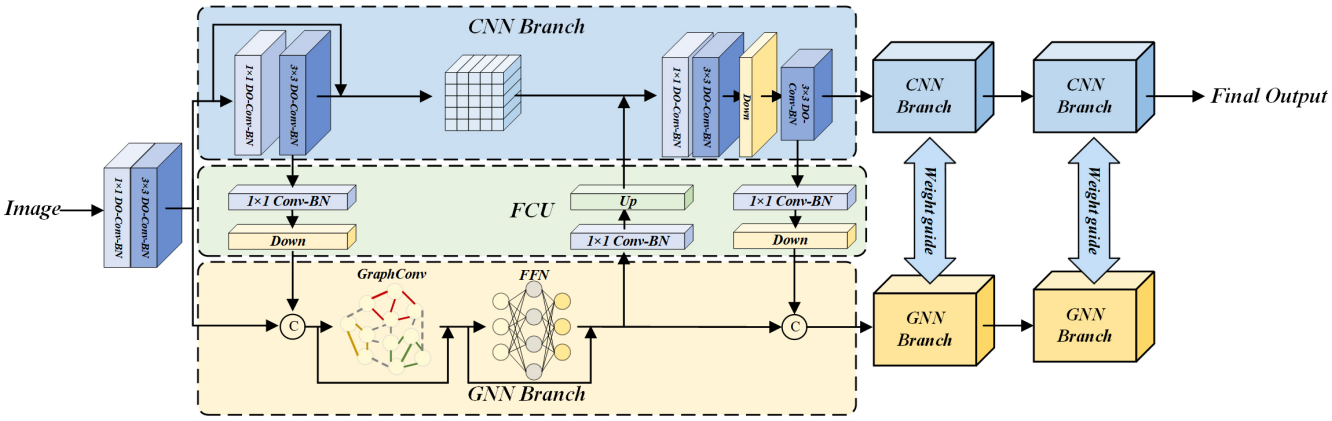

Fig. 3. Congraph's coupling process diagram. It includes CNN branch, GNN branch and FCU.

Algorithm 1 Implementation Process of Our CSAGC Model

Input: I_1, I_2 (*bitemporal image*))
Output: M(*a prediction change mask*)
// *step*1 : *Congraph feature extraction*
for i *in* $\{1, 2\}$ **do**
  $\mathbf{F_1^i}, \mathbf{F_2^i}, \mathbf{F_3^i} = Congraph(\mathbf{I_i})$
end
// *step*2 : *Complementary contextual information*
for i *in* $\{1, 2\}$ **do**
  $\mathbf{F^i_{csam_1}} = CSAM_1(\mathbf{F_1^i}, \mathbf{F_2^i})$
  $\mathbf{F^i_{csam_2}} = CSAM_2(\mathbf{F_1^i}, \mathbf{F_2^i}, \mathbf{F_3^i})$
  $\mathbf{F^i_{csam_3}} = CSAM_3(\mathbf{F_2^i}, \mathbf{F_3^i})$
end
// *step*3 : *Aggregation of differential features*
for j *in* $\{1, 2, 3\}$ **do**
  $\mathbf{F_j} = CAT(\mathbf{F^1_{csam_j}}, \mathbf{F^2_{csam_j}})$
end
// *step*4 : *Information integration*
for j *in* $\{3, 2\}$ **do**
  $\mathbf{F^{j-1}_{sim}} = SIM(Cat(\mathbf{F_{j-1}}, SIM(\mathbf{F_j}))$
end
// *step*5 : *Classification head*
$\mathbf{M} = CH(\mathbf{F^1_{sim}})$

While CNNs have demonstrated their effectiveness in handling complex images, obtaining more detailed global information requires support from a larger RF and improved global modeling capabilities. Therefore, we designed the Congraph hybrid backbone as the encoder, combining CNN and GNN branches to achieve stronger feature representation capabilities. To better visualize the coupling process of Congraph, we have presented the construction of Congraph in Fig. 3. In the GNN branch, we improved upon Vision GNN [50], directly applying graph convolutions and FFN operations on large-scale feature maps. In the CNN branch, we employed DO-Conv [51] as the primary unit, and connected it with the same-sized features from the GNN branch before proceeding with further propagation.

2) GNN Branch: In the GNN branch, for input features of size $\mathbb{R}^{C\times H\times W}$, we first divide it into N patches, transforming each patch into a feature vector $x_i \in \mathbb{R}^D$. We obtain $X = [x_1, x_2, \ldots, x_N]$, where D is the feature dimension. To represent positional information, we add a positional encoding vector to each node feature

$$x_i \longleftarrow x_i + e_i \tag{1}$$

where $e_i \in \mathbb{R}^D$. The features in X are treated as a set of unordered graph nodes, represented as $V = \{v_1, \ldots, v_n\}$. For each node v_i, K nearest neighboring nodes can be identified, and an edge e_{ij} is added from $v_j \in N(v_i)$ to v_i. This way, we obtain a graph $\mathcal{G} = (V, \varepsilon)$, where ε represents all the edges.

For graph-level processing, we start with input features $X \in \mathbb{R}^{N\times D}$. First, we construct a graph based on $\mathcal{G} = G(X)$. Then, we can leverage the graph convolutional layers to aggregate the features of neighboring nodes and exchange information. The specific operations are as follows:

$$\begin{aligned}\mathcal{G}' &= F(\mathcal{G}, W)\\ &= \text{Update}\big(\text{Aggregate}\big(\mathcal{G}, W_{\text{agg}}\big),\ W_{\text{update}}\big)\end{aligned} \tag{2}$$

where $W_{\text{agg}}, W_{\text{update}} \in \mathbb{R}^{N\times D}$ are learnable weights for aggregation and update operations, respectively. The aggregation operation calculates the current node feature by aggregating the features of all neighboring nodes around each node

$$x_i' = h\big(x_i, g\big(x_i, N(x_i), W_{\text{agg}}\big), W_{\text{update}}\big) \tag{3}$$

where $N(x_i)$ is the set of neighbor nodes of x_i. $g(\cdot)$ represents a feature aggregation operation, and $h(\cdot)$ represents a feature update operation. For computational efficiency, the feature aggregation uses max-relative graph convolution [52], which means using a max-pooling feature aggregator without any learnable parameters, to gather the feature differences between x_i and all its neighbors

$$\begin{aligned}g(\cdot) &= x_i'' = \big[x_i, \max\big(\{x_j - x_i\}\big)\big], \quad j\epsilon N(x_i)\\ h(\cdot) &= x_i' = x_i'' W_{\text{update}}.\end{aligned} \tag{4}$$

The processing method of the above graph dimension can be expressed as $X' = \text{GraphConv}(X)$.

After obtaining the aggregated features, we introduce a multihead update operation. The aggregated features x_i'' are divided into h heads, i.e., $\text{head}_1, \text{head}_2, \ldots, \text{head}_h$ and then

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these heads are updated with different weight, respectively. The updates for each head are computed independently and then concatenated to form the final value

$$x_i' = \left[\text{head}_1 W_{\text{update}}^1, \ldots, \text{head}_h W_{\text{update}}^n\right]. \tag{5}$$

Multihead update operation allows the model to update information in multiple representation subspaces.

To prevent the oversmoothing problem that can lead to a decrease in feature discrimination capability in deep GCNs, we introduce a linear layer and a nonlinear layer after the GCN. Together with the GCN, they form the Grapher module. Given the input features $F \in \mathbb{R}^{C\times H\times W}$, the Grapher module can be represented as follows:

$$Y = \sigma(\text{GraphConv}(F W_{\text{in}}))W_{\text{out}} + F \tag{6}$$

where $Y \in \mathbb{R}^{N\times D}$, W_{in} and W_{out} are the weight of fully connected layers, and σ denotes the activation function.

In order to further enhance the feature transformation capability and alleviate oversmoothing, a FFN is used on each node. The FFN module is a simple multilayer perceptron (MLP) with two fully connected layers, represented as

$$Z = \sigma(Y W_1)W_2 + Y \tag{7}$$

where $Z \in \mathbb{R}^{N\times D}$, W_1 and W_2 are the weight of fully connected layers.

A stack Grapher and an FFN module form the a graph unit in the GNN branch.

3) CNN Branch: In the CNN branch, we use N DO-Conv layers to extract local features. DO-Conv is a new way of combining depthwise convolution and traditional convolution. For descriptive purposes, we consider a patch in traditional convolution as a 2-D tensor $\mathbb{P} \in \mathbb{R}^{(M\times N)\times C_{\text{in}}}$, where M and N are the spatial dimensions of the patch, and C_{in} is the number of input feature channels. The trainable kernels of the traditional convolution layer can be represented as a 3-D tensor $W_1 \in \mathbb{R}^{C_{\text{out}}\times(M\times N)\times C_{\text{in}}}$, where C_{out} is the number of output feature channels. In contrast, in depthwise convolution, each C_{in} channel of $\mathbb{P}$ participates in D_{mul} (in DO-Conv $D_{\text{mul}} \geq M \times N$, but in depthwise separable convolution $D_{\text{mul}} < M \times N$.) separate dot products. Each input patch channel (an $M \times N$-dimensional feature) is transformed into a D_{mul}-dimensional feature. D_{mul} is typically referred to as the depth multiplier. Thus, the depthwise convolution kernel can be represented as a 3-D tensor $W_2 \in \mathbb{R}^{D_{\text{mul}}\times(M\times N)\times C_{\text{in}}}$. Since each input channel is transformed into a D_{mul}-dimensional feature, the output of the depthwise convolution operator is a $D_{\text{mul}} \times C_{\text{in}}$-dimensional feature.

DO-Conv is composed of a depthwise convolution with a trainable kernel $\mathbb{D} \in \mathbb{R}^{(M\times N)\times D_{\text{mul}}\times C_{\text{in}}}$ and a traditional convolution with a trainable kernel $\mathbb{W} \in \mathbb{R}^{C_{\text{out}}\times D_{\text{mul}}\times C_{\text{in}}}$, where $D_{\text{mul}} = M \times N$, and the D_{mul} of the DO-Conv used in this article remains constant at this size. For the input patch $\mathbb{P} \in \mathbb{R}^{(M\times N)\times C_{\text{in}}}$, DO-Conv generates a C_{out}-dimensional feature, similar to a convolution layer. In DO-Conv, two operational processes are introduced: feature composition and kernel composition [51]. In this article, we adopt the more computationally efficient kernel composition. We represent

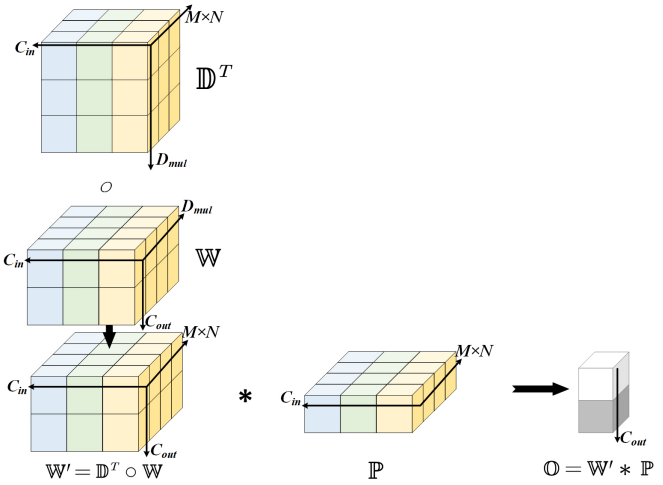

Fig. 4. Describes the kernel composition of DO-Conv, where different colors represent different channels, where ∘ represents depthwise convolution operations, and ∗ represents traditional convolution operations.

the DO-Conv operation using ⊛, so the kernel composition process of DO-Conv can be represented as follows:

$$\mathbb{O} = (\mathbb{D}, \mathbb{W}) \circledast \mathbb{P} = \left(\mathbb{D}^T \circ \mathbb{W}\right) * \mathbb{P} \tag{8}$$

where $*$ means traditional convolution, $\circ$ means depthwise convolution, and $\mathbb{D}^T \in \mathbb{R}^{D_{\text{mul}}\times(M\times N)\times C_{\text{in}}}$ is the transpose of $\mathbb{D}$ on the first and second axes. The use of kernel composition is achieved through $\mathbb{W}' \in \mathbb{R}^{C_{\text{in}}\times(M\times N)\times C_{\text{out}}}$, which combines the operations of $\circ$ and $*$, as depicted in Fig. 4. First, the depthwise convolution operator utilizes the trainable kernel $\mathbb{D}^T$ to transform $\mathbb{W}$, obtaining $\mathbb{W}' = \mathbb{D}^T \circ \mathbb{W}$. Then, a traditional convolution is applied between $\mathbb{W}'$ and $\mathbb{P}$ to obtain $\mathbb{O} = \mathbb{W}' * \mathbb{P}$. The RF of DO-Conv remains $M \times N$ throughout. This can be clearly seen from Fig. 4.

Although DO-Conv slightly increases the number of parameters compared to traditional convolution, its depthwise and pointwise convolutions can be independently calculated, making the model more computationally efficient. Moreover, the separation of depthwise and pointwise convolutions in DO-Conv makes it easier to interpret the model's intermediate results and feature maps, leading to a better understanding of how the model works. Overall, DO-Conv has several advantages such as higher computational efficiency, better generalization performance, and greater interpretability, allowing the CNN branch to achieve better performance with the same computational resources.

4) Feature Coupling: The structure of Congraph is divided into four layers. The first layer performs initial sampling and downsampling to 1/2, while the subsequent three layers follow the structure of a downsampling encoder. In each layer, we employ a feature coupling unit (FCU) to merge the features from the CNN branch and the GNN branch, as shown in Fig. 3. The construction of Congraph is presented in Table I. Initially, the intermediate features from the CNN branch are passed to the GNN branch. After going through a group of Graphers and an FFN, the GNN branch transfers the feature information back to the CNN branch. Following a simple fusion step, the CNN branch continues with the DO-

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TABLE I
LAYER INFORMATION AND STRUCTURE OF THE CONGRAPH ENCODER

stage	output	CNN Branch	FCU	GNN Branch
c1	128×128	3×3, 3→64, strdie=2		
c2	128×128	3×3, C=64, D=3 3×3, C=128, D=3	1×1, C=64 → ← 1×1, C=64	K=9, N=4096, D=3
c3	64×64	3×3, C=128, D=6 3×3, C=256, D=6	1×1, C=128 → ← 1×1, C=128	K=9, N=1024, D=4
c4	32×32	3×3, C=256, D=3 3×3, C=512, D=3	1×1, C=256 → ← 1×1,C=256	K=9, N=256, D=3

Conv blocks. Downsampling operations are performed at the beginning of each stage using stride = 2 convolution. In each layer, the features from the GNN branch are incorporated into the CNN branch and directly serve as the outputs of the encoding stages after passing through a DO-Conv block.

B. Context Spatial Awareness Module

1) Overview: The context information is crucial for the interpretation, classification, and analysis of RS images. It provides a more comprehensive understanding of the image and enables semantic reasoning, thereby improving the performance and reliability of RS image analysis tasks. To effectively aggregate cross-scale information and acquire both detailed and global information, we propose a CSAM. CSAM is an important component connected between the encoder and decoder, as shown in Fig. 5. In CSAM, features from up to three branches are obtained, including the high-level feature branch, low-level feature branch, and current feature branch. We define the operation of CSAM as f, and the representation of the entire module is as follows:

$$F_{\mathrm{CSAM}}^{i}=\begin{cases} f\left(F_E^{i}, F_E^{i+1}\right), & i=1 \\ f\left(F_E^{i-1}, F_E^{i}, F_E^{i+1}\right), & i=2 \\ f\left(F_E^{i-1}, F_E^{i}\right), & i=3 \end{cases} \tag{9}$$

where $F_{\mathrm{CSAM}}^{i} \in \mathbb{R}^{C_i \times H_i \times W_i}$ represents the features output by each stage of CSAM, and F_E^{i-1}, F_E^{i}, F_E^{i-1} represents high-level features, current features, and low-level features, respectively.

2) Current Feature Branch: For the current feature $F_E^{i} \in \mathbb{R}^{C_i \times H_i \times W_i}$, we first input it into a multiscale convolution (MSC) module consisting of dilated convolutions. The MSC module can achieve multiscale RFs, capturing richer feature representations and enhancing the understanding and analysis of complex scenes. It can be represented as follows:

$$F_{\mathrm{DOC}}^{i,j}=\mathrm{DOC}\left(F_E^{i}; K_{3\times3}^{i,j}; d^{j}\right), \quad j \in \{1,2,3,4\} \tag{10}$$

where $F_{\mathrm{DOC}}^{i,j} \in \mathbb{R}^{C_i \times H_i \times W_i}$ represents the output feature, DOC stands for dilated convolution based on DO-Conv. i represents the stage of the encoder, and j represents the level of the dilated convolution. $K_{3\times3}^{i,j}$ represents the kernel size of 3×3, and d^{j} represents the expansion rate. Then, we concatenate four features in the channel dimension to generate F_C. The formula is represented as

$$F_C^{i}=\mathrm{Cat}\left(F_{\mathrm{DOC}}^{i,1}, F_{\mathrm{DOC}}^{i,2}, F_{\mathrm{DOC}}^{i,3}, F_{\mathrm{DOC}}^{i,4}\right). \tag{11}$$

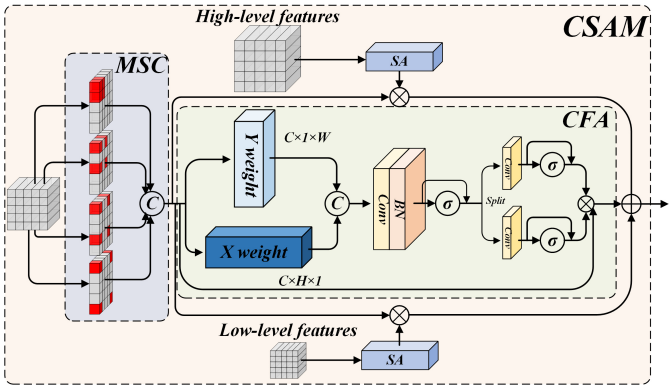

Fig. 5. Construction details of the multiscale mixed attention module are as follows. First, the input features are processed into four different scales of feature maps through a pooling layer, and then input to the multihead cross self-attention (MSCA) module. The larger scale features are responsible for extracting fine-grained features, while the smaller scale features are responsible for extracting global features. The local features and global features are then fused using MSCA, thereby generating multiscale features.

To emphasize the important feature F_C^{i} in the concatenated features, we pass F_C into the current feature attention (CFA) module to update the weights of features in different spatial dimensions, thereby optimizing feature extraction. First, we apply global pooling to transform the features into 1-D feature encodings. Specifically, for a given input $F_C^{i} \in \mathbb{R}^{C_i \times H_i \times W_i}$, we use a pooling kernel of size $(H_i, 1)$ or $(1, W_i)$ to encode each channel along the horizontal or vertical coordinate. Thus, the output of channel c with a height of h can be represented as follows:

$$z_c^{h}(h)=\frac{1}{W}\sum_{0\le i<W} F_C^{i}(h,i). \tag{12}$$

Similarly, the output of the cth channel of width w can be written as

$$z_c^{w}(w)=\frac{1}{H}\sum_{0\le i<H} F_C^{i}(j,w). \tag{13}$$

These two transformations enable the attention module to capture and preserve long-range dependencies and precise spatial information along a specific spatial direction. This helps the network to accurately locate the regions of interest and improves its ability to identify important features.

Next, the two parts of transformed information are concatenated, and a convolutional transformation function is applied to generate an intermediate feature map $F^{hw} \in \mathbb{R}^{C/r\times(H+W)}$ with spatial information in the horizontal and vertical directions, where r represents the reduction ratio. Then, the feature F^{hw} is split into two separate tensors $F^{h} \in \mathbb{R}^{C/r\times H}$ and $F^{w} \in \mathbb{R}^{C/r\times W}$, which are transformed by two 1×1 convolutions f_h and f_w to have the same number of channels as the input. The formula is expressed as

$$\begin{aligned} g^{h} &= \sigma\left(f_h\left(F^{h}\right)\right) \\ g^{w} &= \sigma\left(f_w\left(F^{w}\right)\right) \end{aligned} \tag{14}$$

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where σ represents the sigmoid function. g^h and g^w will be used as attention weights to generate the final output $F^{i,j}_{\text{cur}}$, defined as follows:

$$F^i_{\text{cur}}(i,j) = F^i_C(x,y) \cdot g^h_c(x) \cdot g^w_c(y). \tag{15}$$

The current-scale features contain the primary feature information, while the neighboring-scale features are used to guide current feature modeling.

3) High-Low-Level Feature Branch: The feature information from the high-low-level feature branches (HLFBs) is used to guide the modeling of the current feature. This operation is achieved by spatial attention to coordinate the current feature information. The first and third layers of the encoder include only one of the HLFB branches, utilizing either the high-level or low-level features to coordinate the current feature, and can be represented as follows:

$$\begin{aligned} F^1_{\text{LF}} &= \text{SA}\big(\text{Up}\big(F^2_E\big)\big) \otimes F^1_C \\ F^3_{\text{HF}} &= \text{SA}\big(\text{Down}\big(F^2_E\big)\big) \otimes F^3_C. \end{aligned} \tag{16}$$

The operation Up$(\cdot)$ performs $2\times$ upsampling using bilinear interpolation, while Down$(\cdot)$ performs $2\times$ downsampling using max pooling, SA stands for Spatial Attention. F^i represent different features generated by the encoder in different stages. With this branch structure, the positional information from the low-level features is introduced into F^1_E, and the more detailed alignment information from the high-level features is incorporated into F^3_E.

The second layer utilizes a complete HLFB structure. F^2_E obtains both the detailed alignment information from F^1_C and the positional information from F^3_E, which can be expressed as follows:

$$\begin{aligned} F^2_{\text{LF}} &= \text{SA}\big(\text{Up}\big(F^3_E\big)\big) \otimes F^2_C \\ F^2_{\text{HF}} &= \text{SA}\big(\text{Down}\big(F^1_E\big)\big) \otimes F^2_C. \end{aligned} \tag{17}$$

4) Feature Integration: The weight attention of the current feature branch and the coordination with the HLFB branches are performed in parallel. After effectively coordinating the dual branches, we integrate all the outputs, which can be represented as follows:

$$F^i_{\text{CSAM}} = \begin{cases} F^i_{\text{cur}} \oplus F^i_{\text{LF}} \oplus F^i_E, & i = 1 \\ F^i_{\text{cur}} \oplus \big(F^i_{\text{LF}} \oplus F^i_{\text{HF}}\big) \oplus F^i_E, & i = 2 \\ F^i_{\text{cur}} \oplus F^i_{\text{HF}} \oplus F^i_E, & i = 3 \end{cases} \tag{18}$$

where $\oplus$ denotes element-wise summation. This cross-scale coordination enables the comprehensive fusion of features at different levels, effectively integrating global information with local detail information. It also enhances the perception of positional information, greatly improving the stability and robustness of feature extraction.

C. Spatial Integration Module

The SIM serves as the fundamental unit of our decoder, as shown in Fig. 6. It concatenates the features before CSAM with the current output of SIM to infer the final mask. This process can be represented as follows:

$$F^i_{\text{sim}} = \begin{cases} \text{SIM}\big(F^i_{\text{CSAM}}, \text{TConv}\big(F^{i+1}_{\text{CSAM}}\big)\big), & i = 1,2 \\ \text{SIM}\big(F^i_{\text{CSAM}}\big), & i = 3 \end{cases} \tag{19}$$

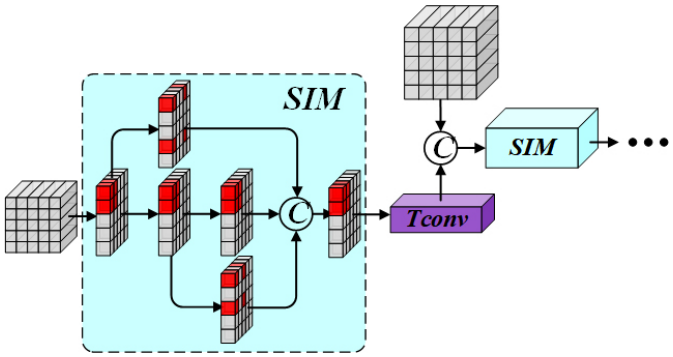

Fig. 6. Construction details of the TR module. It includes three parts: token extractor, TR encoder, and TR decoder.

where F^i_{sim} is the output of SIM layer i, and TConv is a transposed convolutional layer with BN and ReLU.

In SIM, we use dilated convolutions to expand the RF and gather contextual clues from F^i_{CSAM}. Considering that the features from CSAM have different scales, we set different dilation rates for the two layers of dilated convolutions in SIM^1, SIM^2, and SIM^3. Specifically, the dilation rates are set as 3/2, 3/2, and 5/3, respectively. The dilated convolution branch is then concatenated with the cascaded convolutional layers to aggregate these features. The output of each layer in SIM can be represented as follows:

$$F^i_{\text{sim}} = \text{Conv}\Big(\text{Cat}\Big(F^{i,1}_{\text{sim}}, F^{i,2}_{\text{sim}}, F^{i,3}_{\text{sim}}\Big); W^i_{3\times 3}\Big). \tag{20}$$

The SIM module relies on the detailed feature information obtained from CSAM and further scans different-sized regions, which effectively captures object changes in complex RS images.

D. Other Network Details

1) Classification Head: We add two 3×3 convolutional operations at the end of the decoder for channel transformation, resulting in $\mathbb{R}^{2\times H\times W}$.

2) Loss Function: In the CD method, its essence is similar to binary classification tasks in semantic segmentation. Therefore, we use the minimization of cross-entropy (CE) loss to optimize the model parameters. The loss function is defined as follows:

$$L = \frac{1}{H_0 \times W_0} \sum_{h=1,w=1}^{H,W} l(P_{hw}, Y_{hw}) \tag{21}$$

where $l(P_{hw}, y) = -\log(P_{hwy})$ is the CE loss, P_{hw} and Y_{hw} is the label for the pixel at location (h, w).

IV. Experimental Results and Analysis

A. Experimental Setup

1) Evaluation Metrics: To evaluate the effectiveness of our method, we used five evaluation metrics in the experiments: Precision (Pre), Recall (Re), F1-score (F1), intersection over union (IoU), and overall accuracy (OA). Among these metrics, F1 and IoU are the primary evaluation indicators, where larger

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values indicate better model performance. The representations of these evaluation metrics are as follows:

$$\begin{aligned} \text{Pre} &= \frac{\text{TP}}{\text{TP}+\text{FP}} \\ \text{Re} &= \frac{\text{TP}}{\text{TP}+\text{FN}} \\ \text{F1} &= 2\frac{\text{Pre}\cdot\text{Re}}{\text{Pre}+\text{Re}} \\ \text{IoU} &= \frac{\text{TP}}{\text{TP}+\text{FN}+\text{FP}} \\ \text{OA} &= \frac{\text{TP}+\text{pn}}{\text{TP}+\text{TN}+\text{FN}+\text{FP}} \end{aligned} \tag{22}$$

where TP represents true positive, TN represents true negative, FP represents false positive, and FN represents false negative.

2) Implementation Details: Our DL model was implemented using the PyTorch framework. We conducted the training on an Ubuntu system with a Titan RTX GPU and developed the model in the torch1.11 environment. To optimize the model, we employed the stochastic gradient descent (SGD) algorithm with momentum. The momentum parameter was set to 0.99, and the weight decay was set to 0.0005. The learning rate was set to 0.01 for all three datasets. During the training process, we performed validation using the validation set after each training epoch and saved the best-performing model. Subsequently, we tested the model and evaluated its performance. By considering these implementation details, we were able to leverage the computational power of the GPU, utilize the SGD optimization algorithm for model training, and assess the performance and generalization ability of the model through validation and testing datasets.

B. Datasets

We use three high-resolution building CD datasets, namely LEVIR-CD, WHU-CD, and GZ-CD.

LEVIR-CD [25] consists of image pairs from Google Earth, with a total of 637 pairs. Each image pair has a high spatial resolution of 0.5 m/pixel, with an image size of 1024 × 1024 pixels. The dataset covers various types of buildings. We crop the images into nonoverlapping blocks of size 256 × 256 pixels while maintaining the original training, validation, and test partitions. The training set contains 7120 images, the validation set contains 1024 images, and the testing set contains 2048 images.

WHU-CD [53] is a large-scale public CD dataset that consists of a pair of high-resolution images. The image size is 32 507 × 15 354 pixels, with a spatial resolution of 0.075 m/pixel. To conduct experiments, we cropped the WHU-CD dataset into samples of size 256 × 256 pixels for training, validation, and testing. Specifically, the dataset consists of 6096 samples for training, 762 samples for validation, and 760 samples for testing. By performing such cropping and partitioning, we are able to train and evaluate the model using smaller sample sizes while maintaining the integrity of the dataset.

GZ-CD [54] comprises 19 pairs of high-resolution Google Earth image pairs, with a spatial resolution of 0.55 m/pixel. It captures the urban changes in the suburban areas of Guangzhou, China over the course of nearly a decade. The dataset focuses specifically on changes related to buildings. The image sizes vary, ranging from 1006 × 1168 to 4936 × 5224 pixels. To facilitate experiments, we preprocess the GZ-CD dataset by cropping it into nonoverlapping image blocks of size 256 × 256 pixels. These blocks are then divided into training, validation, and testing sets, consisting of 2834, 400, and 325 image blocks, respectively.

C. Comparative Experiment

1) Comparison With SOTA Methods: In this section, we compare CSACG with several SOTA methods, including three CNN and attention-based methods (IFNet [22], DTCD-SCN [21] and SNUNet [23], DMINet [42], and AERNet [41]) and four transformer-based methods (BIT [29], ChangeFormer [30], CropLand [6], and ICIF-Net [55]).

1) DTCDSCN [21] is a method that proposes a dual task constrained deep Siamese convolutional network. It also introduces a dual attention module (DAM) and improves the focus loss to suppress the issue of sample imbalance.
2) IFNet [22] proposes a deep supervised image fusion network for high-resolution dual-temporal RS image CD. It first utilizes a fully convolutional network for feature extraction and then employs a deep supervised difference discrimination network (DDN) for CD. To enhance the integrity of change map boundaries and the compactness of its interior, the network incorporates multilevel deep features and image difference map features through an attention mechanism for fusion.
3) SNUNet [23] inspired by DenseNet and NestedUNet, designs a densely connected ensemble network for CD. With dense skip connections between the encoder and decoder, as well as between decoders, it maintains high-resolution and fine-grained representations. Additionally, it proposes an ensemble channel attention module (ECAM) to refine the most representative features at different hierarchical levels, which is used for final classification.
4) MSCANet [6] is a feature-level method based on TRs. It introduces a CNN-transformer network with multiscale context aggregation to achieve efficient and effective CD. By combining the strengths of CNN and TRs, it can capture rich image features and enhance contextual modeling capabilities.
5) BIT [29] is a feature-level method based on TRs. It extracts tokens with rich image features by incorporating semantic tokens, and then utilizes TRs to enhance their contextual modeling abilities. It combines the advantages of TRs and semantic tokens for CD.
6) ChangeFormer [30] is a pure transformer-based Siamese network. The backbone of this method consists entirely of transformer encoders, without the use of CNN. It employs MLP as the decoder, unifying the architecture within the transformer framework for CD.
7) ICIF-Net [55] proposes a network that combines CNN and TRs to address the challenge of modeling long-range dependencies in CD of RS images. It leverages the

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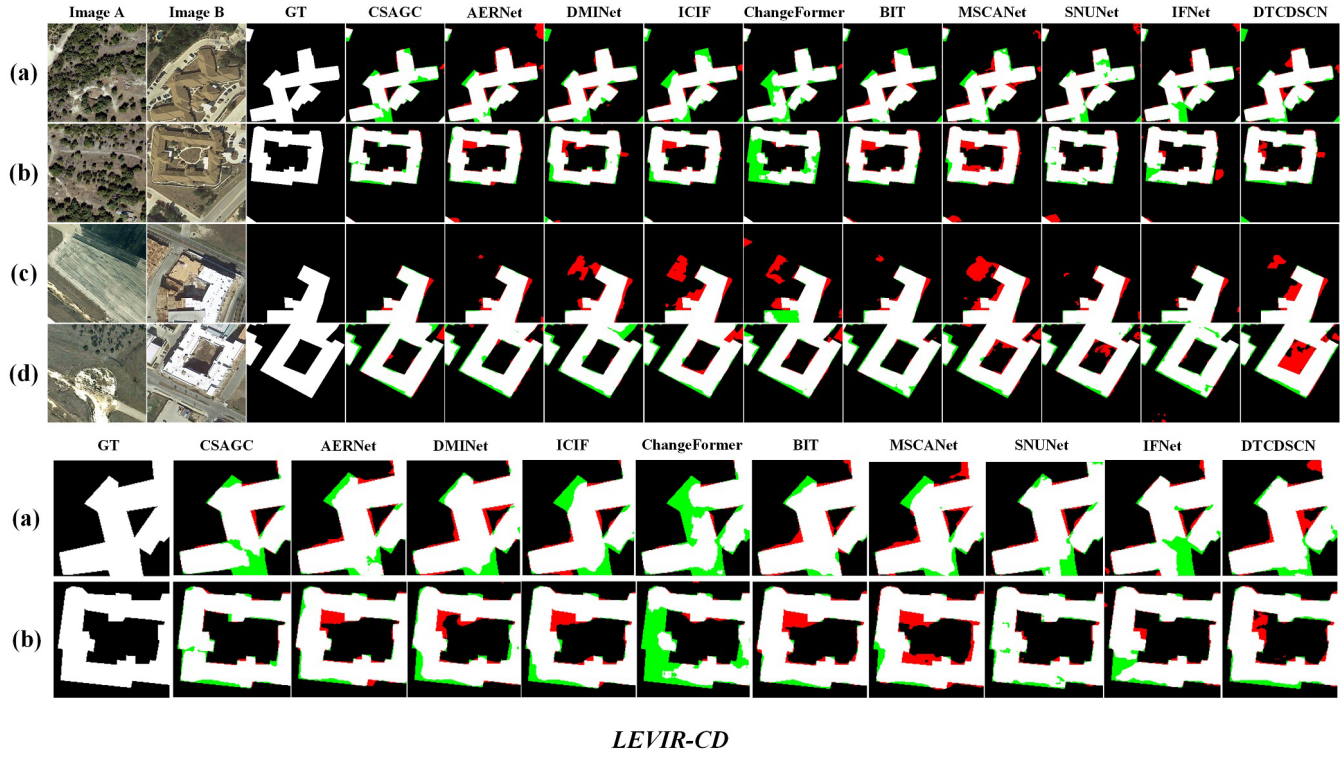

Fig. 7. Visualization results on the LEVIR-CD dataset. Different colors represent different results, with white representing true positive, black representing true negative, red representing false positive, and green representing false negative. (a)–(d) Visualization outcomes for each respective method applied to various images.

strengths of both CNN and TRs for effective modeling of intrascale interactions and interscale feature fusion.

8) DMINet [42] combines self-attention and cross-attention in a single module, introducing an intertemporal joint-attention (JoinAtt) to guide the global feature distribution for each input. Their approach is based on a straightforward backbone with no complex structure, utilizing subtraction and concatenation for multilevel difference aggregation.
9) AERNet [41] introduces an attention-guided edge refinement network (AERNet) that utilizes a global context feature aggregation module (GCFAM) to aggregate information from extracted multilevel context features. It enhances the network's perception and refinement of change regions by combining attention decoding blocks (ADBs) and edge refinement modules (ERMs).

We present the comparative experimental results on the LEVIR-CD dataset in Table II. The red numbers indicate the highest values for each metric, while the blue numbers represent the second highest values. LEVIR-CD is one of the most authoritative datasets for CD tasks. From the table, it can be observed that our CSAGC achieves significant performance improvement compared to the SOTA CD methods. In terms of F1 and IoU metrics, we achieve improvements of 0.75% and 1.22% over the second highest values, respectively. While the IFNet method achieves a high precision of 94.02%, the recall is comparatively lower than other methods, emphasizing the importance of considering the F1 score, which combines precision and recall. We visualize the results on the LEVIR-CD dataset in Fig. 7. From the figure, it is evident that our method performs better in handling object boundaries, as seen in (b) and (d), where our method exhibits fewer false and missed alarms near the object edges. Additionally, our method demonstrates less noise and better recognition of object features, as shown in (c) where CSAGC has fewer nonobject markings. We provide more detailed visualizations corresponding to (a) and (b) in the bottom section of Fig. 7.

TABLE II

COMPARATIVE EXPERIMENTAL RESULTS ON THE LEVIR-CD DATASET. ALL SCORES ARE DESCRIBED AS PERCENTAGES (%). THE HIGHEST VALUES ARE DENOTED IN RED, AND THE SECOND HIGHEST VALUES ARE DENOTED IN BLUE

LEVIR-CD					
Model	Pre	Re	F1	IoU	OA
DTCDSCN	88.53	86.83	87.67	78.05	98.77
DSIFN	**94.02**	82.93	88.13	78.77	98.87
SNUNet	89.18	87.17	88.16	78.83	98.82
BIT	89.24	89.37	89.31	80.68	98.92
MSCANet	89.79	87.57	88.67	79.64	98.86
ChangeFormer	92.05	88.8	90.40	82.48	99.04
ICIF-Net	91.13	90.57	**91.18**	**83.85**	**99.12**
DMINet	92.52	89.95	90.71	82.99	99.07
AERNet	89.97	**91.59**	90.78	83.11	99.07
CSAGC	**92.73**	**91.15**	**91.93**	**85.07**	**99.19**

In Table III, we present the comparative experimental results on the WHU-CD dataset. The WHU-CD dataset has the highest spatial resolution (0.075 m/pixel) among the datasets used

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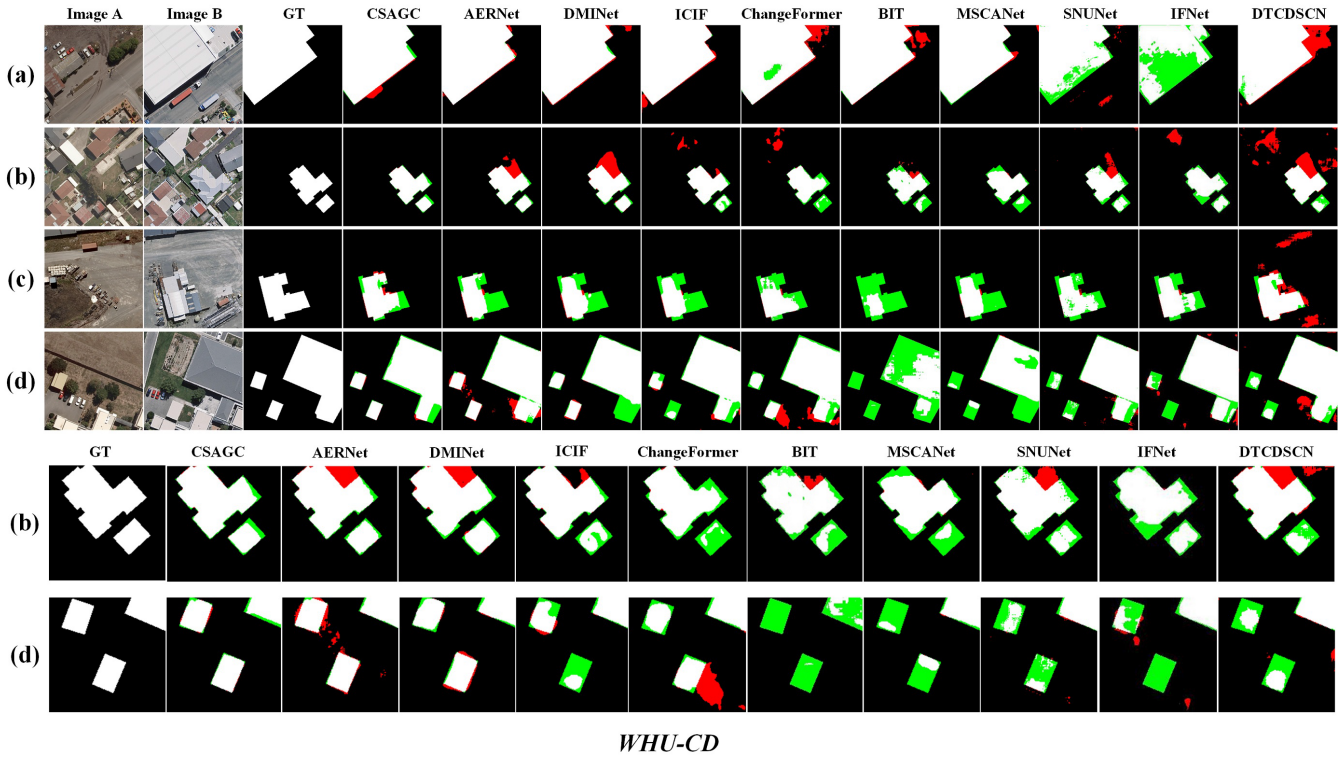

Fig. 8. Visualization results on the WHU-CD dataset. Different colors represent different results, where white represents true positives, black represents true negatives, red represents false positives, and green represents false negatives. (a)–(d) Visualization outcomes for each respective method applied to various images.

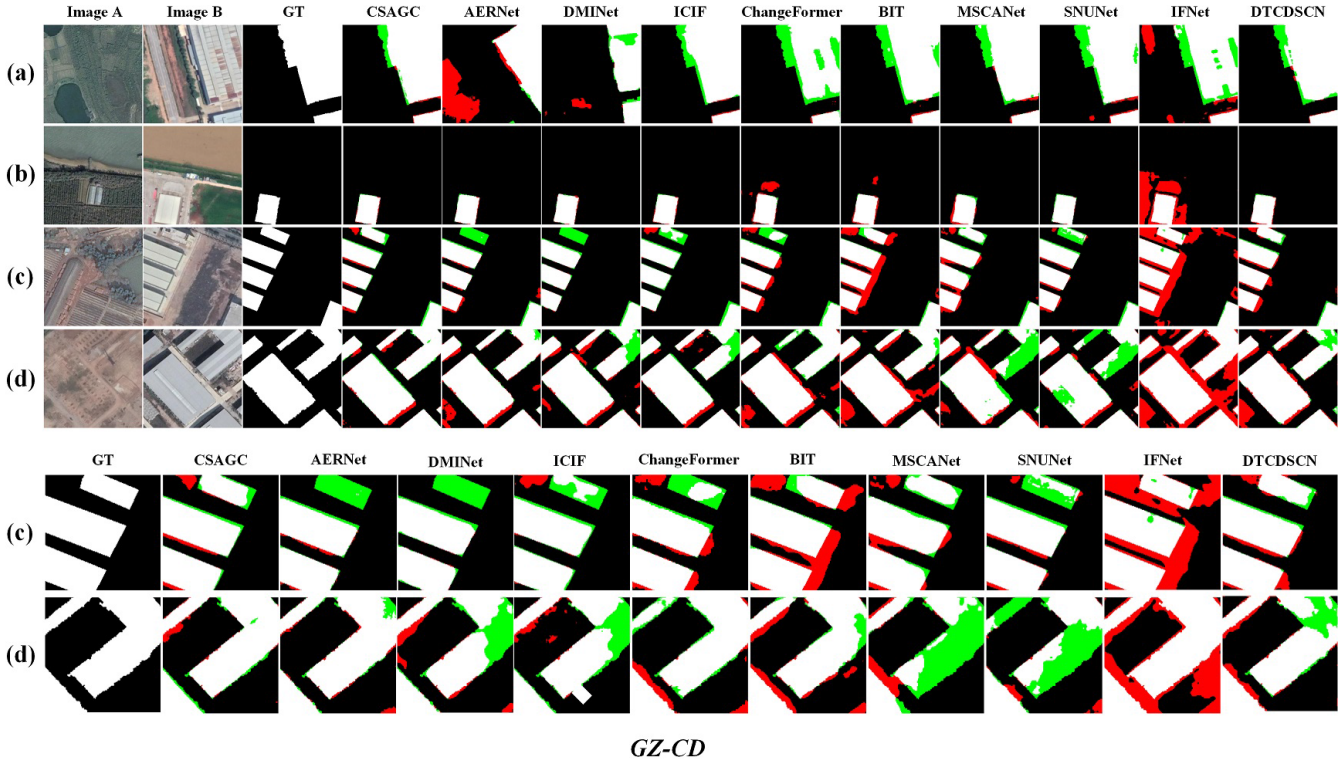

Fig. 9. Visualization results on the GZ-CD dataset. Different colors represent different results, where white represents true positives, black represents true negatives, red represents false positives, and green represents false negatives. (a)–(d) Visualization outcomes for each respective method applied to various images.

in our experiments, which leads to richer feature information in RS images. Consequently, methods with powerful feature extraction capabilities tend to achieve more prominent scores. From the data in the table, it is evident that CSAGC shows significant improvements compared to other methods. Similarly, DTCDSCN, SNUNet, and ICIF-Net, which utilize deep feature

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TABLE III

COMPARATIVE EXPERIMENTAL RESULTS ON THE WHU-CD DATASET. ALL SCORES ARE DESCRIBED AS PERCENTAGES (%). THE HIGHEST VALUES ARE DENOTED IN RED, AND THE SECOND HIGHEST VALUES ARE DENOTED IN BLUE

WHU-CD					
Model	Pre	Re	F1	IoU	OA
DTCDSCN	91.84	89.16	90.48	82.62	99.09
DSIFN	**96.91**	73.20	83.41	71.53	98.83
SNUNet	91.34	85.53	88.34	79.11	98.91
BIT	88.71	86.27	87.47	77.73	98.81
MSCANet	93.16	84.50	88.62	79.56	98.95
ChangeFormer	88.50	85.33	86.88	76.81	98.76
ICIF-Net	92.83	88.70	90.77	83.09	99.13
DMINet	92.65	90.35	91.49	84.31	99.19
AERNet	92.47	**91.89**	**92.18**	**85.49**	**99.25**
CSAGC	**93.64**	**93.57**	**93.61**	**87.98**	**99.38**

TABLE IV

COMPARATIVE EXPERIMENTAL RESULTS ON THE GZ-CD DATASET. ALL SCORES ARE DESCRIBED AS PERCENTAGES (%). THE HIGHEST VALUES ARE DENOTED IN RED, AND THE SECOND HIGHEST VALUES ARE DENOTED IN BLUE

GZ-CD					
Model	Pre	Re	F1	IoU	OA
DTCDSCN	88.19	78.38	83.00	70.93	96.92
DSIFN	55.70	67.41	67.00	43.99	91.74
SNUNet	87.55	80.05	83.64	71.87	97.00
BIT	82.40	78.18	80.23	66.99	96.31
MSCANet	83.04	78.42	80.66	67.59	96.40
ChangeFormer	84.59	65.23	73.66	58.30	95.53
ICIF-Net	**89.90**	80.76	**85.09**	**74.05**	**97.29**
DMINet	87.92	76.79	81.98	69.46	96.77
AERNet	88.06	**81.07**	84.42	73.03	97.13
CSAGC	**88.81**	**82.97**	**85.79**	**75.12**	**97.37**

TABLE V

EFFICIENCY COMPARISON OF EACH METHOD

			LEVIR-CD		WHU-CD		GZ-CD	
	Params(M)	FLOPs(G)	F1	IoU	F1	IoU	F1	IoU
DTCDSCN	41.07	13.21	87.67	78.05	85.03	73.96	83.00	70.93
DSIFN	50.71	82.35	88.13	78.77	74.23	59.03	67.00	43.99
SNUNet	12.03	54.88	88.13	78.83	85.26	74.31	84.25	72.79
BIT	3.55	10.59	89.31	80.68	84.90	73.75	80.23	66.99
MSCANet	16.42	14.77	88.67	79.64	79.64	66.17	80.66	67.59
ChangeFormer	41.03	202.87	90.40	82.48	81.82	69.24	73.66	58.30
ICIF-Net	25.83	25.27	91.18	83.85	90.77	83.09	85.09	74.05
DMINet	6.24	14.55	90.71	82.99	91.49	84.31	81.98	69.46
AERNet	25.36	12.82	90.78	83.11	92.18	85.49	84.42	73.03
CSAGC	50.39	23.11	91.93	85.07	93.61	87.98	85.79	75.12

extraction backbones, also demonstrate better performance compared to other methods. However, CSAGC outperforms them by incorporating multiple contextual information and capturing larger RFs. In comparison to the second highest values in F1 and IoU, CSAGC achieves improvements of 1.43% and 2.49%, respectively. Additionally, we provide visualizations of the results on the WHU-CD dataset in Fig. 8. It is evident from the figure that CSAGC maintains its advantage in handling image details, particularly in the zoomed-in views of (b) and (d).

In Table IV, we present the comparative experimental results on the GZ-CD dataset. The GZ-CD dataset has a slightly lower resolution compared to the other two datasets, resulting in generally lower scores for all methods. Additionally, due to the presence of blurry features, the performance differences among the compared methods are more pronounced in the test results. CSAGC demonstrates good performance on this dataset, with improvements of 0.70% and 1.07% in F1 and IoU, respectively, compared to the second highest values. In the visualizations shown in Fig. 9, our method performs well overall, exhibiting fewer red and green parts. In summary, CSAGC benefits from a powerful Congraph encoder, as well as the effective integration of contextual and spatial information through CSAM and SIM. This enables the extraction of desirable feature information and the establishment of global contextual relationships in complex RS images. As a result, CSAGC demonstrates excellent performance on all three datasets, as evident from both the evaluation metrics and visualized results.

2) Efficiency Comparison: In the efficiency comparison, we evaluate the efficiency of various methods using parameters (Params) and floating-point operations (FLOPs). We set the input size to 256×256 for all methods, and the test results are shown in Table V. In terms of parameter count, our method, as a deep network, has a relatively high number of parameters. However, in terms of experimental results, our deep network performs exceptionally well. Compared to other methods with similar parameter counts, such as DTCDSCN (41.97 M), DSIFN (50.71 M), and Changeformer (41.03 M), CSAGC outperforms them by a significant margin on all three datasets. Despite being a deep network, we have managed to achieve lower FLOPs compared to many other deep networks, such as the multilayer interaction SNUNet (54.88 G), DSIFN (82.35 G), and Changeformer (202.87 G), while still achieving better results.

3) Discussion of SOTA Methods: Through the comparison of method performance and efficiency, our CSAGC exhibits several advantages and limitations. In general, our method aims for superior performance, which is why we employ larger models, resulting in more parameters and computational costs. However, this also delivers better performance. Conversely, methods with lower parameter counts, such as BIT and DMINet, have more room for improvement. AERNet uses a method similar to DSIFN but excels in controlling parameters and computational costs while achieving superior performance. Changeformer, despite not having an advantage in terms of parameters and computational costs, contributes significantly as a pure transformer-based method, exploring the value of TRs in CD tasks. ICIF-Net demonstrates that the combination of CNN and TRs also yields good performance. Clearly, each method has its unique focus, with CSAGC emphasizing performance.

D. Ablation Experiments

1) Encoder Ablation Experiment: To highlight our optimization of Congraph, we have compared the differences between using traditional convolution and DO-Conv, as shown in Table VI.

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TABLE VI
ABLATION EXPERIMENTS FOR THE CONGRAPH MODULE

Module	Params(M)	FLOPs(G)	Tra-Conv	DO-Conv	LEVIR-CD		WHU-CD		GZ-CD	
					F1	IoU	F1	IoU	F1	IoU
Congraph	49.39	88.40	✓	×	91.82	84.94	93.54	87.91	85.78	75.12
Congraph	50.39	23.11	×	✓	91.93	85.07	93.61	87.98	85.79	75.12

TABLE VII
ABLATION EXPERIMENTS INSIDE CSAM

Module	MSC	SA	CFA	LEVIR-CD		WHU-CD		GZ-CD	
				F1	IoU	F1	IoU	F1	IoU
CSAM	×	✓	✓	91.71	84.76	93.24	87.32	84.67	73.36
CSAM	✓	×	✓	91.62	85.54	93.39	86.98	85.41	74.54
CSAM	✓	✓	×	91.24	83.91	93.26	87.35	85.02	73.87
CSAM	✓	✓	✓	91.93	85.07	93.61	87.98	85.79	75.12

TABLE VIII
ABLATION EXPERIMENTS FOR THE OVERALL ARCHITECTURE OF CSAGC

Model	Congraph	CSAM	SIM	LEVIR-CD		WHU-CD		GZ-CD	
				F1	IoU	F1	IoU	F1	IoU
CSAGC	×	✓	✓	91.41	84.18	92.52	86.09	85.48	74.65
CSAGC	✓	×	✓	91.67	84.62	93.14	87.16	84.57	73.27
CSAGC	✓	✓	×	91.64	84.58	92.69	86.38	85.52	74.71
CSAGC	✓	×	×	91.33	84.04	93.06	86.97	84.54	73.67
CSAGC	×	✓	×	91.15	83.78	92.67	86.41	85.12	74.13
CSAGC	×	×	✓	91.23	83.88	92.97	86.79	84.56	73.30
CSAGC (Base)	×	×	×	90.65	82.91	92.14	85.43	84.35	72.98
CSAGC	✓	✓	✓	91.93	85.07	93.61	87.98	85.79	75.12

After replacing DO-Conv, there was a slight decrease in feature performance, but the overall performance remained good. Considering that one of the main advantages of DO-Conv is its lower computational cost, the introduction of DO-Conv with a slight performance improvement under reduced computational load is considered a successful integration.

2) CSAM Internal Ablation Experiment: We divided CSAM into three parts: MSC, spatial attention (SA), and CFA, as shown in Table VII.

MSC plays a crucial role in enhancing CSAM's capacity to capture global information. Ablation experiments, revealing that the performance decline is less pronounced on the LEVIR-CD and WHU-CD datasets compared to the GZ-CD dataset when MSC is removed, underscore the heightened significance of MSC in extracting global information on datasets with lower spatial resolution.

SA module can coordinate multiscale feature information, effectively utilizing the weights brought by the spatial information at different scales, and updating the target weights. Ablation experiments by removing SA resulted in a minor drop in performance on all three datasets. Its presence is essential for better coordinating information at different scales.

CFA enhances the current features in the channel dimension. Ablation experiments by removing CFA resulted in performance drops on all three datasets. CFA serves as the primary information extraction module in CSAM, allowing for the extraction of key features, so its reduction in performance is expected.

In the ablation experiments within CSAM, the performance did not show a significant drop when modules were added or removed. This may be attributed to the fact that the Congraph encoder serves as a feature extractor with a combination of global and local information, providing comprehensive feature extraction. SA and CFA modules further refine the weights on already well-captured features, aiding in the extraction of more detailed features. However, this improvement may not be as prominent compared to ablation experiments conducted on a simple encoder. Therefore, we have conducted an ablation experiment on the overall architecture to further investigate the impact of these modules.

3) Overall Architecture Ablation Experiment: In the ablation experiments of Congraph, we removed the GNN branch and retained only the CNN branch. The results after this replacement are shown in Table VI. From the results in the table, it is evident that the CNN encoder without the GNN branch has a decrease in performance in terms of F1 and IoU. This clearly indicates that the GNN branch provides Congraph with additional feature requirements and satisfies a larger RF. Furthermore, we also observed that the decrease in performance is relatively smaller on the more blurred GZ-CD dataset. This may suggest that Congraph's encoder is more adept at handling complex images, which is desirable. In the future development of RS images, high-resolution RS images are likely to become clearer and more complex. Therefore, the performance of this CNN-GNN interactive encoder may be gradually explored and harnessed as the development progresses.

We conducted ablation experiments on the CSAM module by removing it from the CSAGC framework, as shown in Table VIII. The results demonstrate that the removal of CSAM leads to a decrease in F1 and IoU on all three datasets. By observing the extent of the performance drop on the WHU-CD and GZ-CD datasets, we can conclude that CSAM plays a more significant role in datasets with lower clarity. In datasets with higher clarity and more detailed image features, the encoder is already capable of extracting sufficient feature information. Therefore, CSAM provides effective contextual modeling in complementing the feature extraction capacity when it is not fully utilized.

In the ablation experiments on SIM, we removed the dilated convolutions and retained a simple single-stream convolution structure. The results in Table VIII demonstrate the opposite effect compared to CSAM. There is a minimal decrease in scores on the GZ-CD dataset, while a relatively significant decrease is observed on the WHU-CD dataset. From this observation, we can conclude that dilated convolutions are more effective in extracting global information and performing well on clearer images. However, their contribution to overall feature extraction is less significant in images with lower clarity.

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TABLE IX
ABLATION EXPERIMENTS FOR THE LOSS FUNCTION OF CSAGC

Model	mIoU	FocalLoss	combined loss	CE	LEVIR-CD		WHU-CD		GZ-CD	
					F1	IoU	F1	IoU	F1	IoU
CSAGC	✓				90.35	82.4	91.73	84.73	81.54	68.84
CSAGC		✓			84.09	72.55	88.44	79.27	82.18	69.75
CSAGC			✓		90.15	82.07	91.46	84.27	82.23	69.84
CSAGC				✓	91.93	85.07	93.61	87.98	85.79	75.12

To assess the performance enhancement brought about by Congraph on the overall network, we conducted a set of ablation experiments wherein we excluded CSAM and SIM, retaining only the Congraph encoder. The experimental results indicate that retaining only Congraph led to a decrease in performance across all three datasets.

To evaluate the performance enhancement brought about by Congraph in the overall network, we conducted a series of ablation experiments by removing CSAM and SIM, retaining only the Congraph encoder, as shown in Table VIII. The experimental results indicate that by retaining only Congraph, there was a performance decline across all three datasets.

Through these sets of experimental data, it became evident that the improvement in performance with the addition of CSAM or SIM to the Congraph encoder was quite unstable, and in some cases, it did not result in significant improvements. However, the comprehensive CSAGC model we compared with did show a notable enhancement in its data when CSAM and SIM were added. As a result, we conducted separate tests for CSAM and SIM, retaining them individually. To better compare the contributions of each module, we also conducted a baseline test, where all modules were included simultaneously following the ablation experiment steps.

Comparing the experimental results in Table VIII, it was observed that each module contributed to some extent to the overall performance improvement. An occasional decrease in test results after adding modules may have been due to error margins. However, through the separate testing of each module, it was demonstrated that Congraph, CSAM, and SIM each improved the model's performance to varying degrees.

In conclusion, the CSAGC framework consisting of the Congraph encoder, CSAM, and SIM effectively handles complex RS images by capturing detailed feature information and providing a global perspective from different angles. The ablation experiments demonstrate the contribution of each module to the overall network performance and highlight their distinct applicability.

4) Comparison of Loss Function: Regarding the choice of loss function, we conducted a comparative set of experiments. We trained the model using Focal Loss, CE, mIoU loss, and combined loss [56] functions separately, and the final results are presented in Table IX. Ultimately, through comparison, we selected the CE loss function, which yielded the superior performance.

V. CONCLUSION

In this article, we proposed a context spatial awareness RS image CD network based on graph and convolution interaction, dubbed CSAGC. The Congraph encoder, with CNN-GNN interaction, effectively preserves their respective features and shares local details and global information for robust feature extraction. CSAM leverages the current feature branch and the high-low-level feature branch to explore contextual features and improve target content extraction based on spatial location information. SIM aids in capturing multiscale content and works in conjunction with CSAM to enhance the representation of multiscale contextual features. Extensive experiments demonstrated that our method outperforms SOTA approaches on the LEVIR-CD, WHU-CD, and GZ-CD datasets, achieving significant performance improvements and effectively enhancing CD capabilities.

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Xinyang Song receive the bachelor's degree from the School of Science and Information Science, Qingdao Agricultural University, Qingdao, China, in 2020. He is currently pursuing the master's degree with the School of Information and Electronic Engineering, Shandong Technology and Business University, Yantai, Shandong, China.

His research interests include computer graphics, computer vision, and image progressing.

Zhen Hua received the B.S. and M.S. degrees in electrical automation from the Taiyuan University of Technology, Taiyuan, China, in 1989 and 1992, respectively, and the Ph.D. degree in electronic information engineering from the China University of Mining and Technology, Beijing, China, in 2008.

She is currently a Professor with Shandong Technology and Business University, Yantai, China. Her research interests include computer aided geometric design, information visualization, virtual reality, and image processing.

Jinjiang Li received the B.S. and M.S. degrees in computer science from the Taiyuan University of Technology, Taiyuan, China, in 2001 and 2004, respectively, and the Ph.D. degree in computer science from Shandong University, Jinan, China, in 2010.

From 2004 to 2006, he was an Assistant Research Fellow with the Institute of Computer Science and Technology, Peking University, Beijing, China. From 2012 to 2014, he was a Post-Doctoral Fellow with Tsinghua University, Beijing. He is currently a Professor with the School of Computer Science and Technology, Shandong Technology and Business University, Yantai, China. His research interests include image processing, computer graphics, computer vision, and machine learning.

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